Ch 4 Review: Discrete Random Variables

Remember a discrete random variable X is defined by its **point mass function**

$$p_X(a) := P(X = a)$$

or equivalently defined by its **cumulative distribution function**

$$F_X(t) := P(X \le t) = \sum_{a \le t} p_X(a)$$

What do we remember about these objects?

$0 \le p_X(a) = P(X=a) \le 1$

$0 \le F_X(t) = P(X \le t) \le 1$

$F_X(t) = \sum_{a \le t} p_X(a)$ is increasing

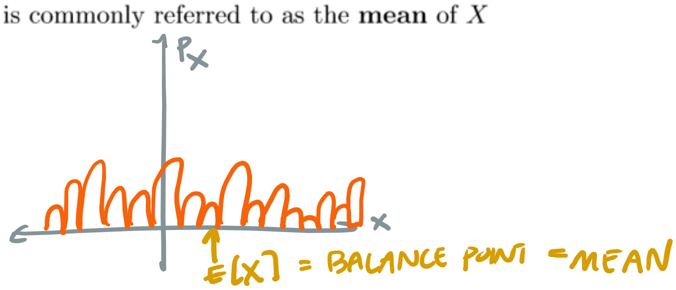
$\lim_{t \to \infty} F_X(t) = P(X \le \infty) = 1$

$\lim_{t \to -\infty} F_X(t) = P(X \le -\infty) = 0$

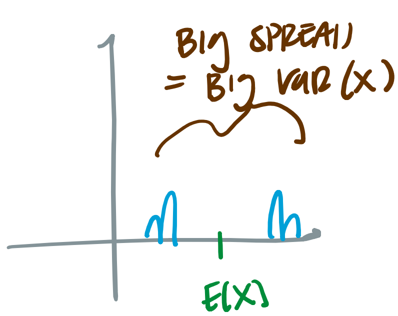
Besides the probability distribution of X, we can also calculate expectations of (functions of) X:

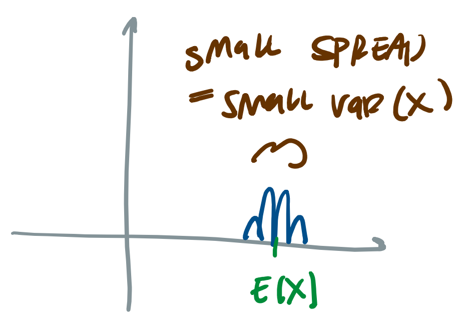
$$E[g(X)] := \sum_a g(a) p_X(a)$$

The value $E[X]$ is commonly referred to as the **mean** of X



The **variance** of X is defined as $Var(X) = E\left[(X - E[X])^2\right] = E[X^2] - (E[X])^2$





A few famous discrete random variables you should be familiar with:

I. Bernoulli

Point Mass Function	Mean	Variance
$p_X(1) = p, p_X(0) = 1 - p$	p	$p(1-p)$

FLIP A COIN   HEADS = 1 "SUCCESS"
TAILS = 0 "FAILS"

$$E[X] = \sum_a a\, p_X(a) = 0 \cdot p_X(0) + 1 \cdot p_X(1)$$
$$= 0 \cdot (1-p) + 1 \cdot p = p$$

$$E[X^2] = \sum_a a^2 \cdot p_X(a) = 0^2 \cdot p_X(0) + 1^2 \cdot p_X(1)$$
$$= 0 \cdot (1-p) + 1 \cdot p = p$$

$$Var(X) = E[X^2] - (E[X])^2 = p - (p)^2 = p - p^2 = p(1-p)$$

II. Binomial

Point Mass Function	Mean	Variance
$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$ $k = 0, 1, \ldots, n$	np	$np(1-p)$

FLIP COIN n TIMES

X = # OF HEADS = "NUMBER OF SUCCESSES"

$P(X=k) = P($k SUCCESSES AND n-k FAILS$)$

$= P($k SUCCESS IN A ROW$) \cdot P($n-k FAILS IN A ROW$) \cdot$ (# OF WAYS TO ORDER k SUCCESS, n-k FAILS)

$= p^k \cdot (1-p)^{n-k} \cdot \binom{n}{k}$

$$1 = \sum_{k=0}^{n} \binom{n}{k} p^k \cdot (1-p)^{n-k} \longleftrightarrow \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k} = (a+b)^n$$

$$E[X] = \sum_{k=0}^{n} k \cdot \binom{n}{k} p^k \cdot (1-p)^{n-k} = \sum_{k=1}^{n} k \cdot \frac{n!}{k!\,(n-k)!} p^k \cdot (1-p)^{n-k}$$

$$= \sum_{k=1}^{n} \frac{n!}{(k-1)!\,(n-k)!} p^k (1-p)^{n-k} = \sum_{k=1}^{n} \frac{n \cdot (n-1)!}{(k-1)!\,((n-1)-(k-1))!} p^k (1-p)^{n-k}$$

$$= n \cdot \sum_{k=1}^{n} \binom{n-1}{k-1} p^k (1-p)^{(n-1)-(k-1)}$$

$j = k-1$

$N = n-1$

$$= np \cdot \underbrace{\sum_{j=0}^{N} \binom{N}{j} p^j \cdot (1-p)^{N-j}}_{1} = np$$

III. Geometric

Point Mass Function	Mean	Variance
$p_X(k) = p(1-p)^{k-1}$ $k = 1, \dots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$

FLIP COIN UNTIL FIRST HEADS

X = # OF attempts

$$\sum_{k=1}^{\infty} p(1-p)^{k-1} = 1 \longrightarrow \sum_{k=0}^{\infty} r^k = \frac{1}{1-r} \text{ FOR } |r|<1$$

$$E[X] = \sum_{k=1}^{\infty} k \cdot p(1-p)^{k-1} = p \cdot \sum_{k=1}^{\infty} k \cdot (1-p)^{k-1}$$

$$= p \cdot \sum_{k=1}^{\infty} \left(k \cdot x^{k-1}\right)\Big|_{x=1-p} = p \cdot \sum_{k=1}^{\infty} \frac{d}{dx}\left(x^k\right)\Big|_{x=1-p}$$

$$= p \cdot \frac{d}{dx}\left(\sum_{k=1}^{\infty} x^k\right)\Big|_{x=1-p} = p \cdot \frac{d}{dx}\left(\frac{x}{1-x}\right)\Big|_{x=1-p}$$

$$= p \cdot \frac{1}{(1-x)^2}\Big|_{x=1-p} = p \cdot \frac{1}{(1-(1-p))^2} = \frac{p}{p^2} = \frac{1}{p}$$

IV. Negative Binomial

Point Mass Function	Mean	Variance
$p_X(k) = \binom{k-1}{r-1}(1-p)^{k-r}p^r$ $k = r, r+1, \ldots$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$

X = # of attempts until rth success

$P(X=k) = (\text{success on } k\text{th}) \cdot (r-1 \text{ successes and } (k-1)-(r-1) \text{ fails in first } k-1 \text{ flips})$

$$= p \cdot \binom{k-1}{r-1} p^{r-1} \cdot (1-p)^{(k-1)-(r-1)}$$

$$1 = \sum_{k=r}^{\infty} \binom{k-1}{r-1} (1-p)^{k-r} \cdot p^r$$

$$E[X] = \sum_{k=r}^{\infty} k \cdot \frac{(k-1)!}{(r-1)!\,(k-r)!} (1-p)^{k-r} \cdot p^r = \sum_{k=r}^{\infty} \frac{k!}{(r-1)!\,(k-r)!} (1-p)^{k-r} \cdot p^r$$

$$= r \cdot \sum_{k=r}^{\infty} \frac{k!}{r!\,(k-r)!} (1-p)^{k-r} \cdot p^r = r \cdot \sum_{k=r}^{\infty} \binom{k}{r} (1-p)^{k-r} \cdot p^r$$

$k = j-1 \longleftrightarrow j = k+1$

$r = R-1 \longleftrightarrow R = r+1$

$$= r \cdot \sum_{j=R}^{\infty} \binom{j-1}{R-1} (1-p)^{(j-1)-(R-1)} p^{R-1}$$

$$= \frac{r}{p} \cdot \underbrace{\sum_{j=R}^{\infty} \binom{j-1}{R-1} (1-p)^{j-R} \cdot p^R}_{1} = \frac{r}{p}$$

V. Poisson

Point Mass Function	Mean	Variance
$p_X(k) = e^{-\lambda}\frac{\lambda^k}{k!}$ $k = 0, 1, \ldots$	λ	λ

$$\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$$

X = ESTIMATOR FOR HOW MANY TIMES INDEPENDENT EVENTS OCCUR IN A FIXED WINDOW OF TIME

$$E[X] = \sum_{k=0}^{\infty} k \cdot e^{-\lambda} \cdot \frac{\lambda^k}{k!} = \sum_{k=1}^{\infty} k \cdot e^{-\lambda} \cdot \frac{\lambda^k}{k!} = \sum_{k=1}^{\infty} e^{-\lambda} \cdot \frac{\lambda^k}{(k-1)!}$$

$j = k-1$

$$= \sum_{j=0}^{\infty} e^{-\lambda} \cdot \frac{\lambda^{j+1}}{j!} = \lambda \cdot \underbrace{\sum_{j=0}^{\infty} e^{-\lambda} \cdot \frac{\lambda^j}{j!}}_{1} = \lambda$$

VI. Hypergeometric

Point Mass Function	Mean	Variance
$p_X(k) = \binom{M}{k}\binom{N-M}{n-k}\binom{N}{n}^{-1}$ $k = 0, 1, \ldots, n$	$n \cdot \frac{M}{N}$	$n\frac{N-n}{N-1} \cdot \frac{M}{N}\left(1 - \frac{M}{N}\right)$

IF AN URN HAS N BALLS, OF WHICH M ARE "WINS" AND YOU PULL n BALLS ONLY, $X = \#$ OF WINS

$$P(X=k) = \frac{\#\text{ OF WAYS TO PULL } k \text{ WINS \& } n-k \text{ LOSSES}}{\#\text{ OF WAYS TO PULL } n \text{ BALLS}}$$

$$= \frac{\binom{M}{k} \cdot \binom{N-M}{n-k}}{\binom{N}{n}}$$

$$\sum_{k=0}^{n} \frac{\binom{M}{k}\binom{N-M}{n-k}}{\binom{N}{n}} = 1$$

$$E[X] = \sum_{k=0}^{n} k \cdot \frac{\binom{M}{k}\binom{N-M}{n-k}}{\binom{N}{n}} = ☹$$